

Prompt Engineering : A $\rightarrow$ Z

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Outline

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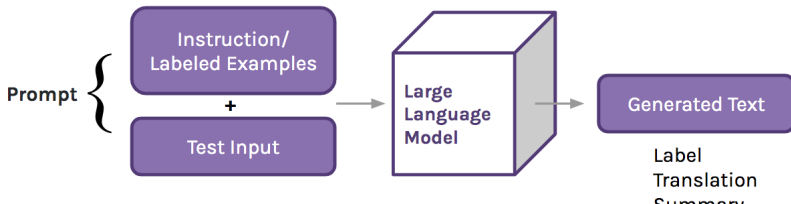
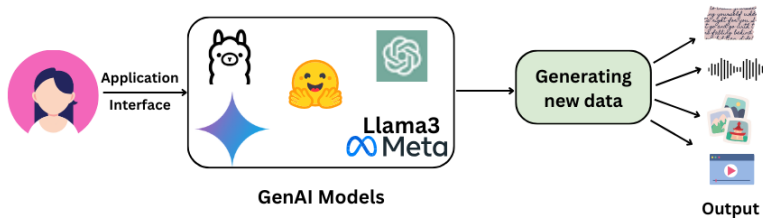
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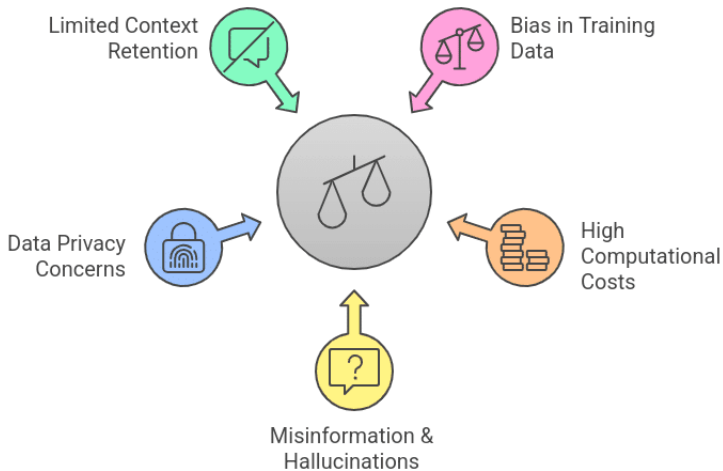
Introduction

Large Language Models



Challenges

Challenges of Large Language Models





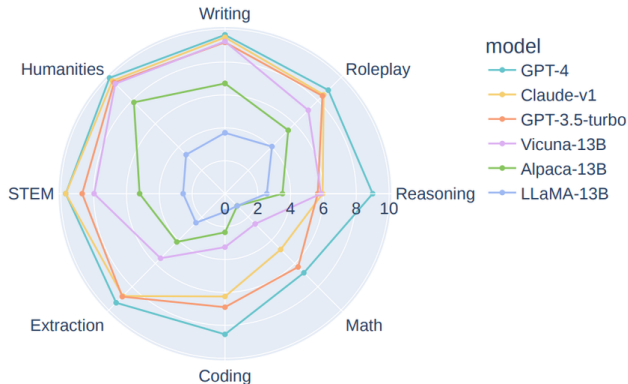
Prompt and Prompt Engineering

Prompt: A message you send to a GenAI

**Prompt Engineering: The “science”
to send the exact right message
to get the output you want**

Why Do We Need Prompt Engineering?

- AI models respond strictly to the **input they receive**.
- Vague prompts lead to **generic or incorrect outputs**.
- Well-designed prompts improve:
 - Accuracy
 - Relevance
 - Consistency
 - Proper tone and format
- Prompt engineering bridges the gap between **human intent** and **AI behavior**.



Prompt Component 1: Role

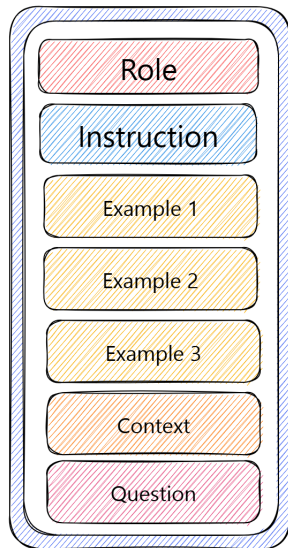
What is a Role?

The role defines **who the AI should act as**. It sets expertise, tone, and perspective.

- Shapes the style and depth of responses
- Helps align AI behavior with user intent
- Reduces generic or unfocused outputs

Example Role

You are an experienced university professor teaching AI to beginners.



Prompt Component 2: Instruction

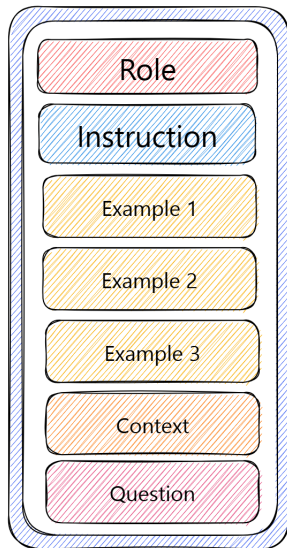
What is an Instruction?

The instruction specifies the **main task** the AI must perform.

- Should be clear, specific, and action-oriented
- Avoids ambiguity and irrelevant responses
- Can include constraints (length, format, tone)

Example Instruction

Explain prompt engineering in simple terms using real-life examples.



Prompt Component 3: Examples (Few-shot)

Why Examples Matter

Examples show the AI **how the output should look**. They guide structure, style, and reasoning.

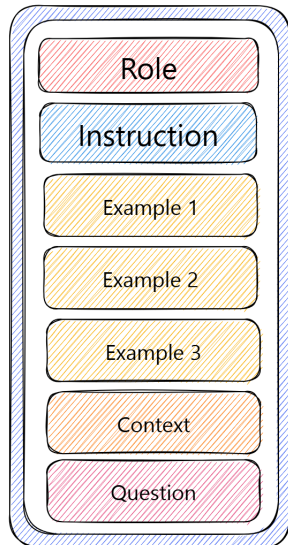
- Improves consistency
- Reduces hallucinations
- Essential for complex or creative tasks

Example

Q: What is AI?

A: AI is the ability of machines to mimic human intelligence.

Q: What is ML?



Prompt Component 4: Context

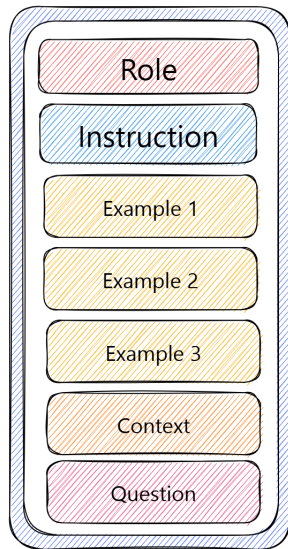
What is Context?

Context provides **background information, constraints, and assumptions** that help the AI tailor its response.

- Narrows down the response space
- Improves relevance and correctness
- Prevents overly generic answers

Example Context

This explanation is for first-year engineering students with no prior knowledge of AI.
Keep the explanation short and intuitive.



Prompt Component 5: Final Question / Answer

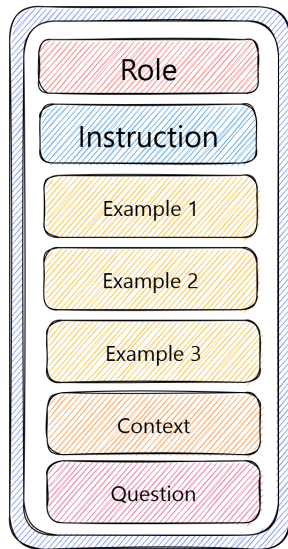
What is the Final Question?

This is the **exact request** you want the AI to respond to. It usually comes at the end of the prompt.

- Directs the AI to produce the final output
- Can specify format, length, or style
- Acts as the execution trigger

Example Final Question

Based on the above context and examples, explain prompt engineering in 5 bullet points using simple language.





Prompt Designs

Designing Prompts for Different Tasks

Tasks covered:

- Text Summarization
- Question Answering
- Text Classification
- Role Playing
- Code Generation
- Reasoning

Prompt Example: Text Summarization

Task

Condense a long text while preserving key ideas.

Prompt

Summarize the following article **in** 5 bullet points.

Focus on the main ideas **and** key conclusions.

Text:

[Paste the article here]

Prompt Example: Question Answering

Task

Answer questions based on given information.

Prompt

Using the information below, answer the question clearly and concisely.

Context:

Photosynthesis is the process by which plants convert sunlight into energy.

Question:

What is photosynthesis?

Prompt Example: Text Classification

Task

Assign a category or label to a piece of text.

Prompt

Classify the following review as
Positive, Negative, or Neutral.

Review:

"The product works fine, but delivery was slow."

Prompt Example: Role Playing

Task

Make the AI behave as a specific persona.

Prompt

You are a career counselor.

Advise a final-year engineering student who **is** confused between higher studies **and** taking up a job.

Prompt Example: Code Generation

Task

Generate executable code from a description.

Prompt

```
Write a Python function to check whether  
a given number is prime.  
Include comments in the code.
```

Prompt Example: Reasoning

Task

Solve a problem using logical, step-by-step reasoning.

Prompt

Solve the problem step by step **and** explain your reasoning.

Problem:

If a shirt costs \$20 after a 20% discount, what was the original price?

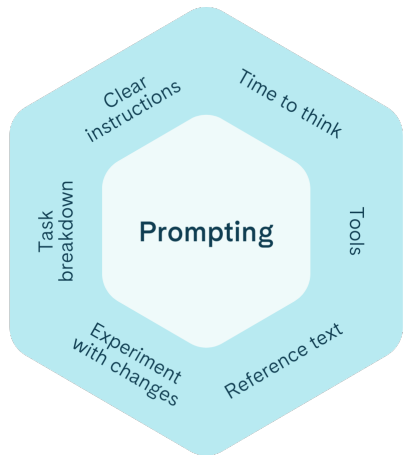


Advanced Prompting Techniques

Advanced Prompting Techniques

"Quality of the prompt determines the usefulness and clarity of the AI's response."

- Anyone can craft effective prompts.
- Advanced prompting sets intent and context.
- Goal: reliable and controllable AI outputs.



Overview of Techniques

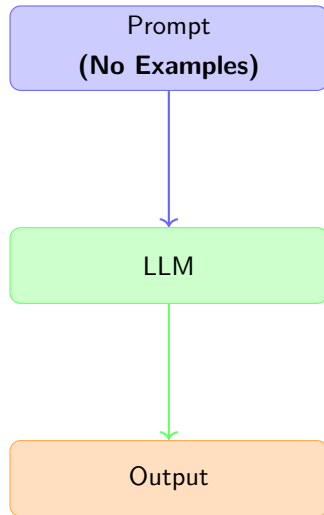
- Zero-shot Prompting
- Few-shot Prompting
- Chain-of-Thought
- Meta Prompting
- Self-Consistency
- Prompt Chaining
- Tree of Thoughts
- ReAct Prompting
- Reflexion
- Directional Stimulus Prompting
- Multimodal CoT
- Retrieval Augmented Generation

Zero-shot Prompting

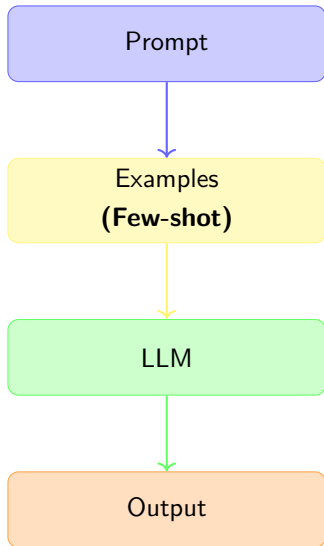
Concept

Ask the AI to perform a task without examples.

Summarize the following article [in](#) 100 words.



Few-shot Prompting



Concept

Provide a few examples and ask the AI to continue similarly.

Examples:

- Save water, save life.
- Reduce waste, reuse today.

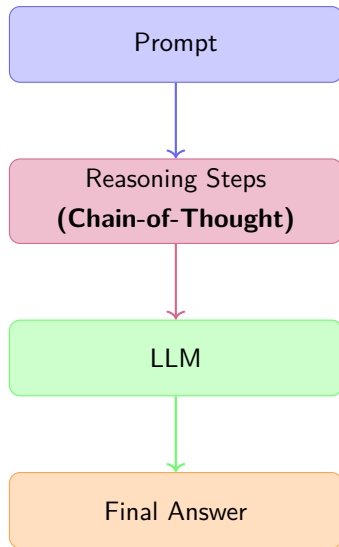
Now create 3 slogans **for** an eco-friendly product.

Chain-of-Thought Prompting

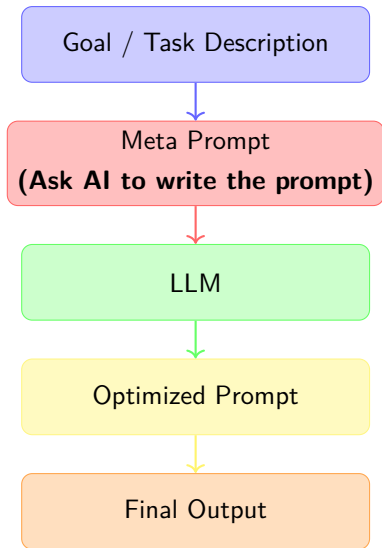
Concept

Encourage step-by-step reasoning before final output.

Solve the problem step by step.
Explain your reasoning clearly.



Meta Prompting



Concept

Ask AI how to design the best possible prompt.

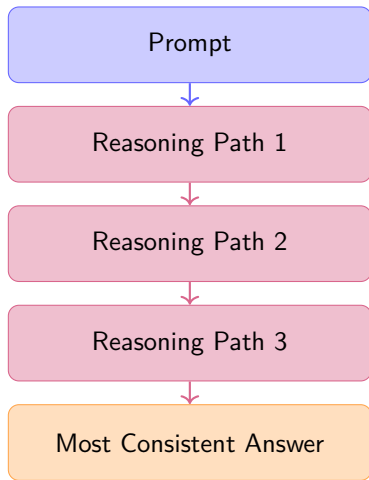
Suggest the best prompt to generate high-quality biology quiz questions.

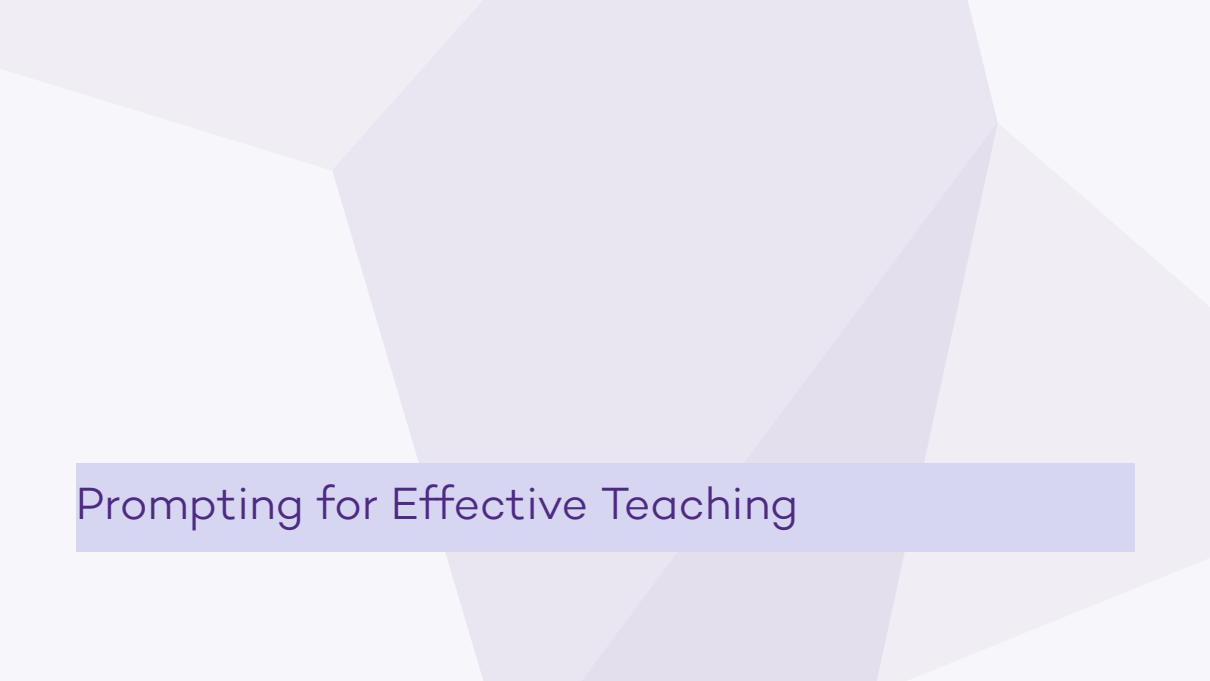
Self-Consistency Prompting

Concept

Ask the AI to generate multiple reasoning paths and select the most consistent or reliable answer.

Solve the problem three different ways.
Show each reasoning path clearly.
Select the answer that appears
most consistent across **all** solutions.
Problem:
A bat **and** a ball cost \$1.10 **in** total.
The bat costs \$1 more than the ball.
How much does the ball cost?





Prompting for Effective Teaching

Stage 1: Ideation

Goal: Generate engaging teaching ideas

Prompt

Act as an experienced science teacher. Suggest 5 creative ways to introduce Photosynthesis to Grade 7 students. Include real-life connections. Present as a bullet list.

Alternative Prompt

Act as an educational innovator. Provide 3 storytelling methods to explain Photosynthesis with examples.

Stage 2: Preparation

Goal: Build conceptual clarity

Prompt

Act as a biology expert. Explain Photosynthesis in simple terms for Grade 7 students. Break into key concepts with examples.

Structured Prompt

Act as a teacher trainer. List prerequisite knowledge for Photosynthesis in a table format with reasons.

Stage 3: Course Planning

Goal: Structure learning

Prompt

Act as a curriculum designer. Create a 3-day lesson plan on Photosynthesis including objectives, activities, and assessments. Present as a table.

Advanced Prompt

Act as an instructional designer. Design a lesson aligned with Blooms taxonomy for Photosynthesis.

Stage 4: Slides & Notes

Goal: Create teaching materials

Prompt

Act as a presentation expert. Create a 10-slide outline on Photosynthesis with bullet points.

With Notes

Act as a teacher. Generate speaker notes explaining each slide clearly.

Stage 5: Activity Design

Goal: Active learning

Prompt

Act as an activity designer. Create a hands-on activity demonstrating Photosynthesis. Include materials, steps, and outcomes.

Gamified Prompt

Act as a game designer. Develop a group simulation activity for Photosynthesis with rules and scoring.

Stage 6: Quiz

Goal: Quick assessment

Prompt

Act as an assessment expert. Create 10 MCQs on Photosynthesis with answers and explanations.

Exit Ticket

Act as a teacher. Generate 5 quick questions to assess understanding at the end of the lesson.

Stage 7: Question Paper

Goal: Summative evaluation

Prompt

Act as an exam setter. Create a 50-mark question paper on Photosynthesis with varied question types and answer key.

Higher Order Thinking

Include case-based and application-oriented questions.

Stage 8: Evaluation

Goal: Measure learning

Prompt

Act as an education expert. Create a rubric to evaluate understanding of Photosynthesis with criteria and performance levels.

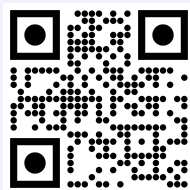
Reflection

Act as a mentor. Suggest reflective questions for teachers after the lesson.

Thank You!

For your attention and participation !!!

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